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Patent Public Search | Text View

United States Patent Application Publication

20250275660

Kind Code

A1

Publication Date

September 04, 2025

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MOBILE CLEANING ROBOT SUSPENSION

Abstract

A mobile cleaning robot movable within an environment can include a body, a drive arm, a container, a biasing element, and a link. The drive arm can be connected to the body and can be movable with respect to the body. The drive arm can support a drive wheel. The container can be connectable to the body and can be configured to carry a fluid therein. The biasing element can be connected to the drive arm to bias the drive wheel toward a floor surface. The link can be pivotably connected to the body and can be connected to the biasing element. The link can be engageable with the tank to adjust the biasing element based on an amount of the fluid in the container.

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Family ID: 87575899

Appl. No.: 19/081264

Filed: March 17, 2025

Related U.S. Application Data

parent US continuation 17878953 20220802 parent-grant-document US 12251059 child US 19081264

Publication Classification

Int. Cl.: A47L11/24 (20060101); A47L11/28 (20060101); A47L11/40 (20060101)

U.S. Cl.:

CPC A47L11/24 (20130101); A47L11/4063 (20130101); A47L11/28 (20130101);
A47L11/4088 (20130101); A47L2201/00 (20130101)

Background/Summary

PRIORITY APPLICATIONS [0001] This is a continuation of and claims priority to U.S. patent application Ser. No. 17/878,953, filed Aug. 2, 2022, the contents of which are incorporated herein by reference in its entirety.

BACKGROUND

[0002] Mobile robots include mobile cleaning robots that can perform cleaning tasks within an environment, such as a home. A mobile cleaning robot can navigate across a floor surface and avoid obstacles while spraying fluid or applying fluid via a pad. The fluid can then be absorbed by a pad to effectively perform mopping operations within the environment as the robot traverses the environment.

SUMMARY

[0003] Mobile cleaning robots can autonomously navigate through environments to perform cleaning operations, often traversing over, and navigating around, obstacles. Mobile cleaning robots include suspension systems to provide sufficient wheel downforce to overcome obstacles and to provide effective cleaning on various surfaces. Because obstacles can vary in shape and size and because floor types can also vary, a required wheel downforce can vary during operation of the robot. Many robots include suspension systems using an extension or compression spring directly connected to wheel arms, which can effectively deliver downforce; however, in mopping robots, the mass or weight of the robot can vary throughout a mission, altering the preferred delivered downforce for optimal cleaning performance.

[0004] This disclosure describes devices and methods that can help to address this problem such as by including a suspension system including a container for storing cleaning fluid that is connected to one or more links and a biasing element connected to a drive arm of the wheel. As the fluid level in the tank changes, the weight or mass of the tank (and the robot) will change affecting the desired downforce for optimal cleaning performance and mobility. The tank can be movable as the volume of fluid changes to move the link, which can, in turn, move the extension spring to adjust a downforce provided to the drive arm and the drive wheel (or wheels) of the robot. In this way, the robot can include a passive suspension adjustment system to help adjust the delivered downforce based on an amount of fluid (and mass of fluid) within the tank to improve cleaning performance and mobility as the amount of fluid in the tank varies over the course of a cleaning mission.

[0005] For example, a mobile cleaning robot movable within an environment can include a body, a drive arm, a container, a biasing element, and a link. The drive arm can be connected to the body and can be movable with respect to the body. The drive arm can support a drive wheel. The container can be connectable to the body and can be configured to carry a fluid therein. The biasing element can be connected to the drive arm to bias the drive wheel toward a floor surface. The link can be pivotably connected to the body and can be connected to the biasing element. The link can be engageable with the tank to adjust the biasing element based on an amount of the fluid in the container.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] In the drawings, which are not necessarily drawn to scale, like numerals may describe similar components in different views. Like numerals having different letter suffixes may represent different instances of similar components. The drawings illustrate generally, by way of example, but not by way of limitation, various embodiments discussed in the present document.

[0007] FIG. 1 illustrates a plan view of a mobile cleaning robot in an environment.

[0008] FIG. 2A illustrates a bottom view of a mobile cleaning robot.

[0009] FIG. 2B illustrates an isometric view of a mobile cleaning robot.

[0010] FIG. 2C illustrates a top view of a mobile cleaning robot.

[0011] FIG. 3A illustrates a cross-section view of a mobile cleaning robot across indicators 3-3 of FIG. 2A in a first condition.

[0012] FIG. 3B illustrates a cross-section view of a mobile cleaning robot across indicators 3-3 of FIG. 2A in a second condition.

[0013] FIG. 4 illustrates a cross-section view of a mobile cleaning robot.

[0014] FIG. 5 illustrates a schematic of a mobile cleaning robot network.

[0015] FIG. 6 illustrates a schematic of a system.

DETAILED DESCRIPTION

Robot Overview

[0016] FIG. 1 illustrates a plan view of a mobile cleaning robot **100** in an environment **40**. The environment **40** can be a dwelling, such as a home or an apartment, and can include rooms **42a-42e**. Obstacles, such as a bed **44**, a table **46**, and an island **48** can be located in the rooms **42** of the environment. Each of the rooms **42a-42e** can have a floor surface **50a-50e**, respectively. Some rooms, such as the room **42d**, can include a rug, such as a rug **52**. The floor surfaces **50** can be of one or more types such as hardwood, ceramic, low-pile carpet, medium-pile carpet, long (or high)-pile carpet, stone, or the like.

[0017] The mobile cleaning robot **100** can be operated, such as by a user **60**, to autonomously clean the environment **40** in a room-by-room fashion. In some examples, the robot **100** can clean the floor surface **50a** of one room, such as the room **42a**, before moving to the next room, such as the room **42d**, to clean the surface of the room **42d**. Different rooms can have different types of floor surfaces. For example, the room **42e** (which can be a kitchen) can have a hard floor surface, such as wood or ceramic tile, and the room **42a** (which can be a bedroom) can have a carpet surface, such as a medium pile carpet. Other rooms, such as the room **42d** (which can be a dining room) can include multiple surfaces where the rug **52** is located within the room **42d**. The robot **100** can be configured to navigate over various floor types through one or more components such as a suspension. The suspension of the robot can allow the robot **100** to navigate over obstacles, such as thresholds between rooms or over rugs, such as the rug **52**.

[0018] Also, during cleaning or traveling operations, the robot **100** can use data collected from various sensors (such as optical sensors) and calculations (such as odometry and obstacle detection) to develop a map of the environment **40**. Once the map is created, the user **60** can define rooms or zones (such as the rooms **42**) within the map. The map can be presentable to the user **60** on a user interface, such as a mobile device, where the user **60** can direct or change cleaning preferences, for example.

[0019] Also, during operation, the robot **100** can detect surface types within each of the rooms **42**, which can be stored in the robot or another device. The robot **100** can update the map (or data related thereto) such as to include or account for surface types of the floor surfaces **50a-50e** of each of the respective rooms **42** of the environment. In some examples, the map can be updated to show the different surface types such as within each of the rooms **42**.

Components of the Robot

[0020] FIG. 2A illustrates a bottom view of the mobile cleaning robot **100**. FIG. 2B illustrates a bottom view of the mobile cleaning robot **100**. FIG. 2C illustrates a top view of the mobile cleaning robot **100**. FIGS. 2A-2C are discussed together below.

[0021] The cleaning robot **100** can be a mobile cleaning robot that can autonomously traverse the floor surface **50** while mopping up dirt or debris **75** from different parts of the floor surface **50**. As depicted in FIGS. 2A-2C, the robot **100** can include a body **102** movable across the floor surface **50**. The body **102** can include multiple connected structures to which movable components of the cleaning robot **100** can be mounted. The connected structures can include an outer housing **103** to

cover internal components of the cleaning robot **100**, a chassis to which drive wheels **104a** and **104b**, a cleaning pad **106** are mounted, and a bumper **108** mounted to the outer housing.

[0022] As shown in FIG. 2A, the robot **100** can include a drive system including actuators **110a** and **110b**, e.g., motors, operable with drive wheels **104a** and **104b**. The actuators **110a** and **110b** can be mounted in the body **102** and can be operably connected to the drive wheels **104a** and **104b**, which are rotatably mounted to the body **102**. The drive wheels **104a** and **104b** can support the body **102** above the floor surface **50**. The actuators **110a** and **110b**, when driven, can rotate the drive wheels **104a** and **104b** to enable the robot **100** to move across the floor surface **50**.

[0023] The controller (or processor) **112** can be located within the housing **103** and can be a programable controller, such as a single or multi-board computer, a direct digital controller (DDC), a programable logic controller (PLC), or the like. In other examples the controller **112** can be any computing device, such as a handheld computer, for example, a smart phone, a tablet, a laptop, a desktop computer, or any other computing device including a processor and communication capabilities. Memory **114** can be one or more types of memory, such as volatile or non-volatile memory, read-only memory (ROM), random-access memory (RAM), magnetic disk storage media, optical storage media, flash-memory devices, and other storage devices and media. The memory **114** can be located within the housing **103** and can be connected to the controller **112** and accessible by the controller **112**.

[0024] As shown in FIG. 2B, the robot **100** can also include a nozzle or ejection port **116** configured to spray or discharge fluid *f* from the robot and onto the floor surface **50**. The nozzle or ejection port **116** can be connected to a pump **118** located within the body **102** or housing **103**. The nozzle or ejection port **116** can be connected to the pump **118** via tubing or piping. The robot **100** can also include, as shown in FIG. 2C, a tank or container **120** configured to store the fluid *f* within the body **102** or housing **103** during a cleaning mission. The pump **118** can be connected to the tank **120** via tubing or piping to connect the nozzle or ejection port **116** to the tank **120**. The pump **118** can also be connected to the controller **112**.

[0025] The controller **112** can operate the actuators **110a** and **110b** to autonomously navigate the robot **100** about the floor surface **50** during a cleaning operation. The actuators **110a** and **110b** are operable to drive the robot **100** in a forward drive direction, in a backwards direction, and to turn the robot **100**. The cleaning pad **106** can help to support a front portion of the body **102** above the floor surface **50**, and the drive wheels **104a** and **104b** support a middle and rear portion of the body **102** above the floor surface **50**. The cleaning pad **106** can be removably mounted to the body **102** of the robot **100**. In this way, the cleaning pad **106** can be user replaceable such as when the cleaning pad **106** becomes dirty during a cleaning mission.

[0026] The control system can further include a sensor system **122** including one or more electrical or optical sensors. The sensor system, as described herein, can include one or more sensor to generate a signal indicative of a current location of the robot **100**, and can include sensors to generate signals indicative of locations of the robot **100** as the robot **100** travels along the floor surface **50**.

[0027] Cliff sensors **124** (shown in FIG. 2A) can be located along a bottom portion of the housing **103**. Each of the cliff sensors **124** can be an optical sensor that can be configured to detect a presence or absence of an object below the optical sensor, such as the floor surface **50**. The cliff sensors **124** can be connected to the controller **112**. The bumper **108** can be removably secured to the body **102** and can be movable relative to body **102** while mounted thereto. In some examples, the bumper **108** form part of the body **102**. Bump sensors **126a** and **126b** (the bump sensors **126**) can be connected to the body **102** and engageable or configured to interact with the bumper **108**. The bump sensors **126** can include break beam sensors, capacitive sensors, switches, or other sensors that can detect contact between the robot **100**, i.e., the bumper **108**, and objects in the environment **40**. The bump sensors **126** can be in communication with the controller **112**.

[0028] An image capture device **128** can be a LIDAR sensor connected to the body **102** and can

extend through the bumper **108** of the robot **100**, such as through an opening of the bumper **108**. The image capture device **128** can be configured to generate a signal based on imagery of the environment **40** of the robot **100** as the robot **100** moves about the floor surface **50**. The image capture device **128** can transmit the signal to the controller **112** for use for navigation and cleaning routines. An image capture device **129** can be a camera connected to the body **102** and can extend through the bumper **108** of the robot **100**. The image capture device **129** can be a camera, such as a front-facing camera, configured to generate a signal based on imagery of the environment **40** of the robot **100** as the robot **100** moves about the floor surface **50**. The image capture device **129** can transmit the signal to the controller **112** for use for navigation and cleaning routines.

[0029] Obstacle following sensors **130** (shown in FIG. 2A) can include an optical sensor facing outward from the bumper **108** and that can be configured to detect the presence or the absence of an object adjacent to a side of the body **102**. The obstacle following sensor **130** can emit an optical beam horizontally in a direction perpendicular (or nearly perpendicular) to the forward drive direction of the robot **100**. The optical emitter can emit an optical beam outward from the robot **100**, e.g., outward in a horizontal direction, and the optical detector detects a reflection of the optical beam that reflects off an object near the robot **100**. The robot **100**, e.g., using the controller **112**, can determine a time of flight of the optical beam and thereby determine a distance between the optical detector and the object, and hence a distance between the robot **100** and the object.

[0030] The robot **100** can also include one or more buttons **132** (or interfaces) that can include a user-operable interface configured to provide commands to the robot, such as to pause a mission, power on, power off, or return to a docking station.

[0031] As shown in FIG. 2B, a lid **134** can be connected to the body **102**, such as over the tank **120**. The lid **134** can be operable to open to access the tank **120** such as for insertion or removal of the tank **120** or for addition of the fluid *f* (e.g., cleaning fluid) to the tank **120**. FIG. 2B also shows that links **136** can be connected to one or more lateral sides of the tank **120**. The **136** can be part of the suspension system as discussed in further detail below.

Operation of the Robot

[0032] In operation of some examples, the robot **100** can be propelled in a forward drive direction or a rearward drive direction. The robot **100** can also be propelled such that the robot **100** turns in place or turns while moving in the forward drive direction or the rearward drive direction.

[0033] When the controller **112** causes the robot **100** to perform a mission, the controller **112** can operate the motors **110** to drive the drive wheels **104** and propel the robot **100** along the floor surface **50**. In addition, the controller **112** can operate the pump **118** to dispense fluid *f* on the floor surface **50**. The controller **112** can execute software stored on the memory **114** to cause the robot **100** to perform various navigational and cleaning behaviors by operating the various motors of the robot **100**.

[0034] The various sensors of the robot **100** can be used to help the robot navigate and clean within the environment **40**. For example, the cliff sensors **124** can detect obstacles such as drop-offs and cliffs below portions of the robot **100** where the cliff sensors **124** are disposed. The cliff sensors **124** can transmit signals to the controller **112** so that the controller **112** can redirect the robot **100** based on signals from the cliff sensors **124**.

[0035] In some examples, the bump sensor **126a** can be used to detect movement of the bumper **108** along a fore-aft axis of the robot **100**. A bump sensor **126b** can also be used to detect movement of the bumper **108** along one or more sides of the robot **100**. The bump sensors **126** can transmit signals to the controller **112** so that the controller **112** can redirect the robot **100** based on signals from the bump sensors **126**.

[0036] The image capture device **128** can be configured to generate a signal based on imagery of the environment **40** of the robot **100** as the robot **100** moves about the floor surface **50**. The image capture device **128** can transmit such a signal to the controller **112**. In some examples, obstacle following sensors **130** can detect detectable objects, including obstacles such as furniture, walls,

persons, and other objects in the environment of the robot **100**. In some implementations, the sensor system can include an obstacle following sensor along a side surface, and the obstacle following sensor can detect the presence or the absence an object adjacent to the side surface. The one or more obstacle following sensors **130** can also serve as obstacle detection sensors, similar to the proximity sensors described herein. The image capture device **129** can be angled in an upward direction, e.g., angled between 5 degrees and 45 degrees from the floor surface **50** about which the robot **100** navigates. The image capture device **129**, when angled upward, can capture images of wall surfaces of the environment so that features corresponding to objects on the wall surfaces can be used for localization.

[0037] The robot **100** can also include sensors for tracking a distance travelled by the robot **100**. For example, the sensor system can include encoders associated with the motors **110** for the drive wheels **104**, and the encoders can track a distance that the robot **100** has travelled. In some implementations, the sensor can include an optical sensor facing downward toward a floor surface. The optical sensor can be positioned to direct light through a bottom surface of the robot **100** toward the floor surface **50**. The optical sensor can detect reflections of the light and can detect a distance travelled by the robot **100** based on changes in floor features as the robot **100** travels along the floor surface **50**.

[0038] The controller **112** can use data collected by the sensors of the sensor system to control navigational behaviors of the robot **100** during the mission. For example, the controller **112** can use the sensor data collected by obstacle detection sensors of the robot **100**, (the cliff sensors **124**, the bump sensors **126**, and the image capture device **128**) to enable the robot **100** to avoid obstacles within the environment of the robot **100** during the mission.

[0039] The sensor data can also be used by the controller **112** for simultaneous localization and mapping (SLAM) techniques in which the controller **112** extracts features of the environment represented by the sensor data and constructs a map of the floor surface **50** of the environment. The sensor data collected by the image capture device **128** can be used for techniques such as vision-based SLAM (VSLAM) in which the controller **112** extracts visual features corresponding to objects in the environment **40** and constructs the map using these visual features. As the controller **112** directs the robot **100** about the floor surface **50** during the mission, the controller **112** can use SLAM techniques to determine a location of the robot **100** within the map by detecting features represented in collected sensor data and comparing the features to previously stored features. The map formed from the sensor data can indicate locations of traversable and non-traversable space within the environment. For example, locations of obstacles can be indicated on the map as non-traversable space, and locations of open floor space can be indicated on the map as traversable space.

[0040] The sensor data collected by any of the sensors can be stored in the memory **114**. In addition, other data generated for the SLAM techniques, including mapping data forming the map, can be stored in the memory **114**. These data produced during the mission can include persistent data that are produced during the mission and that are usable during further missions. In addition to storing the software for causing the robot **100** to perform its behaviors, the memory **114** can store data resulting from processing of the sensor data for access by the controller **112**. For example, the map can be a map that is usable and updateable by the controller **112** of the robot **100** from one mission to another mission to navigate the robot **100** about the floor surface **50**.

[0041] The persistent data, including the persistent map, helps to enable the robot **100** to efficiently clean the floor surface **50**. For example, the map enables the controller **112** to direct the robot **100** toward open floor space and to avoid non-traversable space. In addition, for subsequent missions, the controller **112** can use the map to optimize paths taken during the missions to help plan navigation of the robot **100** through the environment **40**.

Suspension Examples

[0042] FIG. 3A illustrates a cross-section view of a mobile cleaning robot **300** across indicators 3-3

of FIG. 2A in a first condition. FIG. 3A illustrates a cross-section view of the mobile cleaning robot **300** across indicators **3-3** of FIG. 2A in a second condition. FIGS. 3A and 3B are discussed together below. The mobile cleaning robot **300** can be similar to the robot **100** discussed above; like numerals can represent similar components.

[0043] The mobile cleaning robot **300** can include a body **302** and a suspension system **335** including a drive arm **338** connected to the body at an arm pivot **339**. The drive arm **338** can be movable or rotatable with respect to the body **302** about the arm pivot **339**. The drive arm **338** can also be connected to a drive wheel **310** such as to support the drive wheel **310**. The drive wheel **310** engage the floor surface **50** to help move the **300** about the environment **40**. Optionally, the suspension system can include a 4-bar linkage connected to the body **302**.

[0044] The mobile cleaning robot **300** can also include a container or tank **320** connectable to the body **302**. For example, the tank **320** can be locatable or positionable within the **302**. The tank **320** can optionally be removable from the body **302** of the mobile cleaning robot **300**. The tank **320** can be configured to carry a fluid *f* therein such as for dispensing by a sprayer or nozzle (e.g., nozzle **116**), as discussed above. The tank **320** can be any container or tank configured to receive and retain a fluid therein. In other examples the tank or container **320** can be configured to receive dry debris therein.

[0045] The suspension system **335** can also include a biasing element **340** connected to the drive arm **338** and connected to a link **336**. The biasing element **340** can be any biasing element such as an extension spring, compression spring, spring bar, torsion spring, or the like. The biasing element **340** can be connected to a pivot **342** to connect the biasing element **340** to the drive arm **338** to bias the drive wheel **310** toward the floor surface **50**. The biasing element **340** can also be connected to a pivot **344** to connect the biasing element **340** to the link **336** such as to allow rotation of the biasing element **340** with respect to the drive arm **338** and the link **336**.

[0046] The suspension system **335** can also include the link **336** that can be movably (such as one or more of pivotably, rotatably, or slidably) connected to the body **302** at pivot **346**. The link **336** can include a projection **348**, which can be a boss, protrusion, connector, slider, or other feature. The projection **348** can be engaged with a guide **350** of the tank **320** such that the **348** and the guide **350** can form a sliding linkage or rotating mechanism to allow movement of the tank to cause movement of the link **336**, as discussed in further detail below. Though only one link **336** is shown, the mobile cleaning robot **300** can include two (as shown in FIG. 2C) or more links, such as 3, 4, 5, 6, 7, 8, 9, 10, or the like. The link **336** can be engageable or engaged with the tank **320**, such as through the projection **348** and the guide **350** to adjust the biasing element **340** based on an amount of the fluid *f* in the tank **320**. The link can be L-shaped, as shown in FIGS. 3A and 3B to accommodate three connection points, but can have other shapes in other examples, such as an X-shape, a C-shape, a T-shape, an S-shape, an irregular shape, or the like.

[0047] In operation of some examples, the tank **320** can be filled with fluid *f* such that tank **320** is full or nearly full as shown in FIG. 3A. In such a condition, the weight or mass of the fluid *f* can apply a force to the tank **320** which can apply a force to the projection **348** via the guide **350**. This can cause the link **336** to move towards a rear of the robot or to be in its most downward position, causing movement of the pivot **344**. This movement can cause the biasing element **340** to extend to a length *L1* to increase the downforce *F1* applied by the drive wheel **310** to the floor surface **50**. That is, movement of the biasing element **340** due to the mass of the fluid *f* can change a downforce delivered to the drive wheel **310**. Though the link **336** is discussed as moving rearward when the tank **320** is full, the link **336** can be configured to move in any direction.

[0048] As the fluid *f* is used or dispensed by the mobile cleaning robot **300**, such as for cleaning or mopping operations (as discussed above), the fluid level *f* 'can be reduced, as shown in FIG. 3B. This reduction in fluid level *f* 'can reduce a weight or mass of the fluid within the tank **320** allowing upward movement of the **320** (caused by the biasing element **340**), such that the distance *D1* (shown in FIG. 3A) between the tank **320** and the body **302** is reduced to distance *D2* (shown in

FIG. 3B). This movement of the tank **320** within and relative to the body **302** can cause movement of the projection **348**. Movement of the projection **348** within and with the guide **350** can cause movement or rotation of the link **336** about the pivot **346**, such as to allow the link **336** to move to the position **336'** shown in FIG. 3B. This movement causes the pivot **344** to move closer to the pivot **342**, reducing a length of the biasing element **340** to **L2**, which can be shorter than the length **L1**. Because the biasing element **340** can be an extension spring (or a similar biasing element where length impacts exerted force), the shorter length **L2** can cause a smaller force to be applied to the pivot **342** by the biasing element **340**. This, in turn, can cause the downforce **F2** to be smaller than the downforce **F1**, which helps to compensate for the reduction in weight of the tank **320** caused by the reduction in fluid level **f** in the tank **320**.

[0049] In this way, the suspension system **335** of the mobile cleaning robot **300** can help to passively change the downforce provided by the drive wheel **310** to the floor surface **50** based on the amount of fluid **f** (or weight of the fluid **f**) within the tank **320**, which can help to improve mobility of the robot throughout an environment and can help to improve cleaning efficiency of the mobile cleaning robot **300** and effectiveness over a course of a mission.

[0050] Though the suspension **335** is discussed as operating with a tank for storing fluids for mopping, the suspension can also be implemented with a dry tank (such as for vacuuming) or a wet and dry tank. In either instance, the suspension **335** can adjust the downforce of the wheel to be increased as the weight of the bin increases due to accumulation of debris therein.

[0051] FIG. 4 illustrates a cross-section view of a mobile cleaning robot **400**. The mobile cleaning robot **400** can be similar to the robot **100** and the mobile cleaning robot **300** discussed above; the mobile cleaning robot **400** can differ in that its suspension system can provide active control of a length of the biasing element to adjust downforce of the drive wheel. Any of the robots discussed above or below can be modified to include such components.

[0052] The mobile cleaning robot **400** can include a body **402** and a suspension system **435** including a drive arm **438** connected to the body at an arm pivot **439**. The drive arm **438** can be movable or rotatable with respect to the body **402** about the arm pivot **439**. The drive arm **438** can also be connected to a drive wheel **410** such as to support the drive wheel **410**. The drive wheel **410** can be engaged with the floor surface **50** to move the **400** about the environment **40**.

[0053] The mobile cleaning robot **400** can also include a container or tank **420** connectable to the body **402**. For example, the tank **420** can be locatable or positionable within the **402**. The tank **420** can optionally be removable from the body **402** of the mobile cleaning robot **400**. The tank **420** can be configured to carry a fluid **f** therein such as for dispensing by a sprayer or nozzle (e.g., nozzle **116**), as discussed above.

[0054] The suspension system **435** can also include a biasing element **440** connected to the drive arm **438**. The biasing element **440** can be any biasing element such as an extension spring, compression spring, spring bar, torsion spring, or the like. The biasing element **440** can be connected to a pivot **442** to apply a force on the pivot **442** to bias the drive wheel **410** toward the floor surface **50**.

[0055] The suspension system **435** can also include a drive system **452** including a rack **454**, a pinion **456**, and a bearing **458**. The rack **454** can be a geared rack such as a straight rack or a curved rack including teeth, which can be engageable or engaged with the pinion **456**. The pinion **456** can be connected to an actuator or motor **460**, which can be in communication with a controller (e.g., **112**). The motor **460** can be operable to rotate the pinion **456**. The bearing **458** can be connected to the pinion **456** and movable therewith. The bearing **458** can be connected to the biasing element **440** such that the biasing element **440** can be movable with the pinion **456** and the bearing **458**.

[0056] The mobile cleaning robot **400** can also include sensors **462a** and **462b** (sensors **462**) connected to the body **402** and engaged with the tank **420**. One or more of the sensors **462** can be configured to produce a signal based on a weight or mass of the tank **420**, and can be configured

transmit the signal to the controller. One or more of the sensors **462** can be a single point load cell, a digital load cell, a beam load cell, a canister load cell, a hydraulic load cell, a strain gauge, a capacitive load cell, a piezoelectric transducer, or the like. Optionally, the sensors **462** can be one or more break beam sensors that can trigger and un-trigger to have two set-points. In some examples, fluid levels can be determined through fluid level measurements (such as using capacitive, resistive, or magnetic level sensors). Level sensors can be used, such as by a controller to determine a load of the tank.

[0057] In operation of some examples, the tank **420** can be filled with fluid *f* to the level *f1*, such that the tank **420** is full or relatively full, such as at the beginning of a cleaning or mopping mission. The sensors **462** can measure the weight or mass of the tank **420** (or can sense the fluid as it is dispensed) and the fluid *f* therein and can transmit a signal to the controller based on the sensed or detected mass or load. The controller can determine a mass of the tank based on the load signal and can instruct or operate the motor **460** to operate the pinion **456** to rotate to move along the rack **454**. For example, when the controller determines that the fluid level is full or the load is high, the controller can operate the motor **460** to drive the pinion **456** to a far rear portion of the rack **454**, which can move or extend the biasing element **440** to its maximum length, increasing a force applied by the biasing element **440** to the drive arm **438** increasing a downforce applied to the floor surface **50** by the drive wheel **410**.

[0058] As the fluid *f* is used by the mobile cleaning robot **400**, the fluid level can decrease, decreasing the mass or weight of the tank **420**. The change can be sensed by the sensors **462** which can alter the load signal(s) transmitted to the controller by the sensors **462**. The controller can then determine that the weight has lowered (or changed) and can therefore determine that the downforce required to be delivered by the drive wheel **410** is relatively lower. The controller can then operate the motor **460** to drive the pinion **456** to rotate to move the pinion **456** along the rack **454**, such as toward the front of the body **402** to reduce a length of the biasing element **440**. The reduction of length of the biasing element **440** can decrease a force applied to the drive arm **438**, decreasing the downforce applied by the drive wheel **410** to the floor surface **50**.

[0059] In this way, the suspension system **435** of the mobile cleaning robot **400** can be actively controlled by a controller of the mobile cleaning robot **400** to adjust the downforce provided by the drive wheel **410** based on the fluid level within the tank **420**. This active control of the downforce can help to improve mobility of the robot throughout an environment and can help to improve cleaning efficiency and effectiveness of the mobile cleaning robot **400** over a course of a mission.

Network Examples

[0060] FIG. **5** is a diagram illustrating an example of a communication network **500** that can enable networking between the mobile robot **501** and one or more other devices, such as a mobile device **504**, a cloud computing system **506**, or another autonomous robot **508** separate from the mobile robot **501**. Though the network below is discussed as the robot **501** being a primary robot, the robot **508** can be the primary robot.

[0061] Using the communication network **510**, the robot **501**, the mobile device **504**, the robot **508**, and the cloud computing system **506** can communicate with one another to transmit and receive data from one another. In some examples, the robot **501**, the robot **508**, or both the robot **501** and the robot **508** communicate with the mobile device **504** through the cloud computing system **506**. Alternatively, or additionally, the robot **501**, the robot **508**, or both the robot **501** and the robot **508** communicate directly with the mobile device **504**. Various types and combinations of wireless networks (e.g., Bluetooth, radio frequency, optical based, etc.) and network architectures (e.g., mesh networks) can be employed by the communication network **510**.

[0062] In some examples, the mobile device **504** can be a remote device that can be linked to the cloud computing system **506** and can enable a user to provide inputs. The mobile device **504** can include user input elements such as, for example, one or more of a touchscreen display, buttons, a microphone, a mouse, a keyboard, or other devices that respond to inputs provided by the user. The

mobile device **504** can also include immersive media (e.g., virtual reality) with which the user can interact to provide input. The mobile device **504**, in these examples, can be a virtual reality headset or a head-mounted display.

[0063] The user can provide inputs corresponding to commands for the mobile robot **501**. In such cases, the mobile device **504** can transmit a signal to the cloud computing system **506** to cause the cloud computing system **506** to transmit a command signal to the mobile robot **501**. In some implementations, the mobile device **504** can present augmented reality images. In some implementations, the mobile device **504** can be a smart phone, a laptop computer, a tablet computing device, or other mobile device.

[0064] According to some examples discussed herein, the mobile device **504** can include a user interface configured to display a map of the robot environment. A robot path, such as that identified by a coverage planner, can also be displayed on the map. The interface can receive a user instruction to modify the environment map, such as by adding, removing, or otherwise modifying a keep-out zone in the environment; adding, removing, or otherwise modifying a focused cleaning zone in the environment (such as an area that requires repeated cleaning); restricting a robot traversal direction or traversal pattern in a portion of the environment; or adding or changing a cleaning rank, among others.

[0065] In some examples, the communication network **510** can include additional nodes. For example, nodes of the communication network **510** can include additional robots. Also, nodes of the communication network **510** can include network-connected devices that can generate information about the environment **40**. Such a network-connected device can include one or more sensors, such as an acoustic sensor, an image capture system, or other sensor generating signals, to detect characteristics of the environment **40** from which features can be extracted. Network-connected devices can also include home cameras, smart sensors, or the like.

[0066] In the communication network **510**, the wireless links can utilize various communication schemes, protocols, etc., such as, for example, Bluetooth classes, Wi-Fi, Bluetooth-low-energy, also known as BLE, 802.15.4, Worldwide Interoperability for Microwave Access (WiMAX), an infrared channel, satellite band, or the like. In some examples, wireless links can include any cellular network standards used to communicate among mobile devices, including, but not limited to, standards that qualify as 1G, 2G, 3G, 4G, 5G, or the like. The network standards, if utilized, qualify as, for example, one or more generations of mobile telecommunication standards by fulfilling a specification or standards such as the specifications maintained by International Telecommunication Union. For example, the 4G standards can correspond to the International Mobile Telecommunications Advanced (IMT-Advanced) specification. Examples of cellular network standards include AMPS, GSM, GPRS, UMTS, LTE, LTE Advanced, Mobile WiMAX, and WiMAX-Advanced. Cellular network standards can use various channel access methods, e.g., FDMA, TDMA, CDMA, or SDMA.

[0067] FIG. **6** illustrates a block diagram of an example machine **600** upon which any one or more of the techniques (e.g., methodologies) discussed herein may perform. Examples, as described herein, may include, or may operate by, logic or a number of components, or mechanisms in the machine **600**. Circuitry (e.g., processing circuitry) is a collection of circuits implemented in tangible entities of the machine **600** that include hardware (e.g., simple circuits, gates, logic, etc.). Circuitry membership may be flexible over time. Circuitries include members that may, alone or in combination, perform specified operations when operating. In an example, hardware of the circuitry may be immutably designed to carry out a specific operation (e.g., hardwired). In an example, the hardware of the circuitry may include variably connected physical components (e.g., execution units, transistors, simple circuits, etc.) including a machine readable medium physically modified (e.g., magnetically, electrically, moveable placement of invariant massed particles, etc.) to encode instructions of the specific operation. In connecting the physical components, the underlying electrical properties of a hardware constituent are changed, for example, from an

insulator to a conductor or vice versa. The instructions enable embedded hardware (e.g., the execution units or a loading mechanism) to create members of the circuitry in hardware via the variable connections to carry out portions of the specific operation when in operation. Accordingly, in an example, the machine readable medium elements are part of the circuitry or are communicatively coupled to the other components of the circuitry when the device is operating. In an example, any of the physical components may be used in more than one member of more than one circuitry. For example, under operation, execution units may be used in a first circuit of a first circuitry at one point in time and reused by a second circuit in the first circuitry, or by a third circuit in a second circuitry at a different time. Additional examples of these components with respect to the machine **600** follow.

[0068] In alternative embodiments, the machine **600** may operate as a standalone device or may be connected (e.g., networked) to other machines. In a networked deployment, the machine **600** may operate in the capacity of a server machine, a client machine, or both in server-client network environments. In an example, the machine **600** may act as a peer machine in peer-to-peer (P2P) (or other distributed) network environment. The machine **600** may be a personal computer (PC), a tablet PC, a set-top box (STB), a personal digital assistant (PDA), a mobile telephone, a web appliance, a network router, switch or bridge, or any machine capable of executing instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while only a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein, such as cloud computing, software as a service (SaaS), other computer cluster configurations.

[0069] The machine (e.g., computer system) **600** may include a hardware processor **602** (e.g., a central processing unit (CPU), a graphics processing unit (GPU), a hardware processor core, or any combination thereof), a main memory **604**, a static memory (e.g., memory or storage for firmware, microcode, a basic-input-output (BIOS), unified extensible firmware interface (UEFI), etc.) **606**, and mass storage **608** (e.g., hard drive, tape drive, flash storage, or other block devices) some or all of which may communicate with each other via an interlink (e.g., bus) **630**. The machine **600** may further include a display unit **610**, an alphanumeric input device **612** (e.g., a keyboard), and a user interface (UI) navigation device **614** (e.g., a mouse). In an example, the display unit **610**, input device **612** and UI navigation device **614** may be a touch screen display. The machine **600** may additionally include a storage device (e.g., drive unit) **608**, a signal generation device **618** (e.g., a speaker), a network interface device **620**, and one or more sensors **616**, such as a global positioning system (GPS) sensor, compass, accelerometer, or other sensor. The machine **600** may include an output controller **628**, such as a serial (e.g., universal serial bus (USB), parallel, or other wired or wireless (e.g., infrared (IR), near field communication (NFC), etc.) connection to communicate or control one or more peripheral devices (e.g., a printer, card reader, etc.).

[0070] Registers of the processor **602**, the main memory **604**, the static memory **606**, or the mass storage **608** may be, or include, a machine readable medium **622** on which is stored one or more sets of data structures or instructions **624** (e.g., software) embodying or utilized by any one or more of the techniques or functions described herein. The instructions **624** may also reside, completely or at least partially, within any of registers of the processor **602**, the main memory **604**, the static memory **606**, or the mass storage **608** during execution thereof by the machine **600**. In an example, one or any combination of the hardware processor **602**, the main memory **604**, the static memory **606**, or the mass storage **608** may constitute the machine readable media **622**. While the machine readable medium **622** is illustrated as a single medium, the term “machine readable medium” may include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) configured to store the one or more instructions **624**.

[0071] The term “machine readable medium” may include any medium that is capable of storing, encoding, or carrying instructions for execution by the machine **600** and that cause the machine **600**

to perform any one or more of the techniques of the present disclosure, or that is capable of storing, encoding or carrying data structures used by or associated with such instructions. Non-limiting machine readable medium examples may include solid-state memories, optical media, magnetic media, and signals (e.g., radio frequency signals, other photon based signals, sound signals, etc.). In an example, a non-transitory machine readable medium comprises a machine readable medium with a plurality of particles having invariant (e.g., rest) mass, and thus are compositions of matter. Accordingly, non-transitory machine-readable media are machine readable media that do not include transitory propagating signals. Specific examples of non-transitory machine readable media may include: non-volatile memory, such as semiconductor memory devices (e.g., Electrically Programmable Read-Only Memory (EPROM), Electrically Erasable Programmable Read-Only Memory (EEPROM)) and flash memory devices; magnetic disks, such as internal hard disks and removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks.

[0072] The instructions **624** may be further transmitted or received over a communications network **626** using a transmission medium via the network interface device **620** utilizing any one of a number of transfer protocols (e.g., frame relay, internet protocol (IP), transmission control protocol (TCP), user datagram protocol (UDP), hypertext transfer protocol (HTTP), etc.). Example communication networks may include a local area network (LAN), a wide area network (WAN), a packet data network (e.g., the Internet), mobile telephone networks (e.g., cellular networks), Plain Old Telephone (POTS) networks, and wireless data networks (e.g., Institute of Electrical and Electronics Engineers (IEEE) 802.11 family of standards known as Wi-Fi®, IEEE 802.16 family of standards known as WiMax®, IEEE 802.15.4 family of standards, peer-to-peer (P2P) networks, among others. In an example, the network interface device **620** may include one or more physical jacks (e.g., Ethernet, coaxial, or phone jacks) or one or more antennas to connect to the communications network **626**. In an example, the network interface device **620** may include a plurality of antennas to wirelessly communicate using at least one of single-input multiple-output (SIMO), multiple-input multiple-output (MIMO), or multiple-input single-output (MISO) techniques. The term “transmission medium” shall be taken to include any intangible medium that is capable of storing, encoding or carrying instructions for execution by the machine **600**, and includes digital or analog communications signals or other intangible medium to facilitate communication of such software. A transmission medium is a machine readable medium.

NOTES AND EXAMPLES

[0073] The following, non-limiting examples, detail certain aspects of the present subject matter to solve the challenges and provide the benefits discussed herein, among others. [0074] Example 1 is a mobile cleaning robot movable within an environment, the mobile cleaning robot comprising: a body; a drive arm connected to the body and movable with respect to the body, the drive arm supporting a drive wheel; a container connectable to the body and configured to carry a fluid therein; a biasing element connected to the drive arm to bias the drive wheel toward a floor surface; and a link pivotably connected to the body and connected to the biasing element, the link engageable with the container to adjust the biasing element based on an amount of the fluid in the tank. [0075] In Example 2, the subject matter of Example 1 optionally includes wherein movement of the biasing element changes a downforce delivered to the drive wheel. [0076] In Example 3, the subject matter of Example 2 optionally includes wherein the link is connected to the tank through at least one of a sliding and pivoting engagement. [0077] In Example 4, the subject matter of Example 3 optionally includes wherein the link is connected to a first lateral side of the tank. [0078] In Example 5, the subject matter of Example 4 optionally includes a second link connected to a second lateral side of the tank, opposite the link and the first lateral side of the tank; and a second biasing element connected to a second drive arm to bias a second drive wheel connected to the second drive arm toward the floor surface. [0079] In Example 6, the subject matter of any one or more of Examples 1-5 optionally include wherein the biasing element includes an extension spring. [0080] In Example 7, the subject matter of any one or more of Examples 1-6 optionally include

wherein the tank is configured to move vertically based on an amount of fluid within the tank.

[0081] In Example 8, the subject matter of Example 7 optionally includes wherein vertical movement of the tank causes the link to at least one of slide or rotate with respect to the body.

[0082] Example 9 is a mobile cleaning robot movable within an environment, the mobile cleaning robot comprising: a body; a drive wheel arm connected to the body and movable with respect to the body, the drive wheel arm supporting a drive wheel; tank connectable to the body and configured to receive and retain a fluid therein; a biasing element connected to the drive arm to bias the drive wheel toward a floor surface; and an actuator connected to the body and the biasing element; a transducer connected to the tank and configured to produce a load signal based on an amount of the fluid within the tank; and a controller configured to: determine a mass of the tank based on the load signal; and operate the actuator based on the determined mass to adjust the biasing element. [0083]

In Example 10, the subject matter of Example 9 optionally includes wherein the actuator includes a rack connected to the body and the actuator includes a pinion engaged with the rack, the pinion connected to the biasing element and drivable to move the pinion along the rack to adjust a length of the biasing element. [0084] In Example 11, the subject matter of any one or more of Examples 9-10 optionally include wherein movement of the biasing element changes a downforce delivered to by the drive wheel. [0085] In Example 12, the subject matter of Example 11 optionally includes wherein the actuator is connected to a first lateral side of the tank. [0086] In Example 13, the subject matter of Example 12 optionally includes a second actuator connected to a second lateral side of the tank, opposite the actuator and the first lateral side of the tank. [0087] In Example 14, the subject matter of any one or more of Examples 9-13 optionally include wherein the biasing element includes an extension spring and the transducer includes a load cell. [0088] Example 15 is a mobile cleaning robot movable within an environment, the mobile cleaning robot comprising: a body; a drive wheel connected to the body and engageable with a floor surface; a container connectable to the body and configured to receive and retain a mass therein; a biasing element connected to the drive wheel to bias the drive wheel toward a floor surface; and a link connected to the body and the biasing element, the link engageable with the container to adjust a force applied by the drive wheel based on a mass carried in the container. [0089] In Example 16, the subject matter of Example 15 optionally includes wherein movement of the biasing element changes a downforce delivered to the drive wheel. [0090] In Example 17, the subject matter of Example 16 optionally includes wherein the link is connected to the container through at least one of a sliding and pivoting engagement. [0091] In Example 18, the subject matter of Example 17 optionally includes wherein the link is located on a first lateral side of the tank. [0092] In Example 19, the subject matter of Example 18 optionally includes a second link located on a second lateral side of the tank, opposite the link and the first lateral side of the tank; and a second biasing element connected to a second drive arm to bias a second drive wheel connected to the second drive arm toward the floor surface. [0093] In Example 20, the subject matter of any one or more of Examples 15-19 optionally include wherein the tank is configured to move vertically based on an amount of fluid within the container. [0094] In Example 21, the apparatuses or method of any one or any combination of Examples 1-20 can optionally be configured such that all elements or options recited are available to use or select from.

[0095] The above detailed description includes references to the accompanying drawings, which form a part of the detailed description. The drawings show, by way of illustration, specific embodiments in which the invention can be practiced. These embodiments are also referred to herein as “examples.” Such examples can include elements in addition to those shown or described. However, the present inventors also contemplate examples in which only those elements shown or described are provided. Moreover, the present inventors also contemplate examples using any combination or permutation of those elements shown or described (or one or more aspects thereof), either with respect to a particular example (or one or more aspects thereof), or with respect to other examples (or one or more aspects thereof) shown or described herein.

[0096] In the event of inconsistent usages between this document and any documents so incorporated by reference, the usage in this document controls. In this document, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein.” Also, in the following claims, the terms “including” and “comprising” are open-ended, that is, a system, device, article, composition, formulation, or process that includes elements in addition to those listed after such a term in a claim are still deemed to fall within the scope of that claim.

[0097] The above description is intended to be illustrative, and not restrictive. For example, the above-described examples (or one or more aspects thereof) may be used in combination with each other. Other embodiments can be used, such as by one of ordinary skill in the art upon reviewing the above description. The Abstract is provided to comply with 37 C.F.R. § 1.72 (b), to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. Also, in the above Detailed Description, various features may be grouped together to streamline the disclosure. This should not be interpreted as intending that an unclaimed disclosed feature is essential to any claim. Rather, inventive subject matter may lie in less than all features of a particular disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description as examples or embodiments, with each claim standing on its own as a separate embodiment, and it is contemplated that such embodiments can be combined with each other in various combinations or permutations. The scope of the invention should be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.

Claims

1. (canceled)
2. A mobile cleaning robot movable within an environment, the mobile cleaning robot comprising: a body; a drive wheel rotatably connected to the body and operable to move the mobile cleaning robot about the environment; a drive arm connected to the body and supporting the drive wheel; a tank connectable to the body and configured to carry a fluid therein; a biasing element connected to the drive arm to bias the drive wheel toward a floor surface; and a link pivotably connected to the body and connected to the biasing element, the link engageable with the tank by a sliding engagement to adjust the biasing element based on an amount of the fluid in the tank.
3. The mobile cleaning robot of claim 2, wherein the link is connected to a first lateral side of the tank.
4. The mobile cleaning robot of claim 3, further comprising: a second link connected to a second lateral side of the tank, opposite the link and the first lateral side of the tank; and a second biasing element connected to a second drive arm to bias a second drive wheel connected to the second drive arm toward the floor surface.
5. The mobile cleaning robot of claim 2, wherein the biasing element includes an extension spring.
6. The mobile cleaning robot of claim 2, wherein the tank is configured to move vertically based on an amount of fluid within the tank.
7. The mobile cleaning robot of claim 6, wherein vertical movement of the tank causes the link to slide with respect to the body.
8. A mobile cleaning robot movable within an environment, the mobile cleaning robot comprising: a body; a pair of drive wheels rotatably connected to the body and operable to move the mobile cleaning robot about the environment; a pair of drive arms connected to the body and supporting the pair of drive wheels, respectively; a tank connected to the body and configured to carry a fluid therein; a pair of biasing elements connected to the pair of drive arms, respectively, to bias the pair of drive wheels toward a floor surface; and a pair of links each connected to the body and

connected to the pair of biasing elements, respectively, the pair of links each engaged with the tank to adjust a downforce delivered to the pair of drive wheels based on an amount of the fluid in the tank.

9. The mobile cleaning robot of claim 8, wherein the pair of links are separately connected to the tank and include at least one of a sliding and pivoting engagement.

10. The mobile cleaning robot of claim 9, wherein the pair of links include: a first link located on a first lateral side of the tank; and a second link located on a second lateral side of the tank.

11. The mobile cleaning robot of claim 8, wherein the tank is configured to move vertically based on an amount of fluid within the tank.

12. The mobile cleaning robot of claim 11, wherein vertical movement of the tank causes each of the pair of links to slide or rotate with respect to the body.

13. The mobile cleaning robot of claim 8, wherein the tank is removable and replaceable by an end user.

14. A mobile cleaning robot movable within an environment, the mobile cleaning robot comprising: a body; a drive wheel rotatably connected to the body and operable to move the mobile cleaning robot about the environment; a drive arm connected to the body and supporting the drive wheel; a container connectable to the body and configured to carry a fluid therein; a biasing element connected to the drive arm to bias the drive wheel toward a floor surface; and a link pivotably connected to the body and connected to the biasing element, the link engageable with the container to adjust a downforce delivered to the drive wheel based on an amount of the fluid in the container.

15. The mobile cleaning robot of claim 14, wherein the link includes: a projection at least partially engageable with the container.

16. The mobile cleaning robot of claim 15, wherein the container includes: a guide engaged with the projection to at least partially form at least partially for a translating or rotating engagement between the link and the container.

17. The mobile cleaning robot of claim 16, wherein the container is configured to move vertically based on an amount of fluid within the container.

18. The mobile cleaning robot of claim 17, wherein vertical movement of the container causes the link and the projection to slide or rotate with respect to the body to adjust Page 5 the biasing element based on the amount of the fluid in the container and to thereby change the downforce delivered to the drive wheel.

19. The mobile cleaning robot of claim 18, wherein the link is connected to a first lateral side of the container.

20. The mobile cleaning robot of claim 19, further comprising: a second link connected to a second lateral side of the container, opposite the link and the first lateral side of the container; and a second biasing element connected to a second drive arm to bias a second drive wheel connected to the second drive arm toward the floor surface.

21. The mobile cleaning robot of claim 15, wherein the link is L-shaped.
